

# SANGJUN (SAMUEL) Ko

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## EDUCATION

### University of Illinois Urbana-Champaign

Master of Science in Statistics

Bachelor of Science in Statistics (CGPA: 3.92/4.00)

Champaign, Illinois

August 2025 - May 2027 (Expected)

January 2023 - May 2025

- Mathematics Minor
- Coursework: Data Science Programming Methods(STAT447), Advanced Data Analysis(STAT448), Statistical Modeling II(STAT426), Statistical Data Management(STAT440), Statistics and Probability II(STAT410), Elementary Real Analysis(MATH444), Linear Algebra(MATH257)

## WORK EXPERIENCE

### STAT107 Data Science Discovery

Undergraduate Course Assistant

Champaign, Illinois

August 2023 - Present

- Assist over 300 students with fundamental data science concepts through interactive labs
- Manage Git for learning platform modifications, benefiting 1000+ students; create, edit, and proofread course materials
- Collaborate with a team of 100+ staff and 2 professors in weekly meetings

### STAT443 Professional Statistics

Undergraduate Course Assistant

Champaign, Illinois

January 2025 - Present

- Grade assignments and provide feedback to 20+ students on statistical analysis and consulting using Canvas LMS
- Act as a client in mock projects, simulating real-world consulting challenges

### Applied Technologies for Learning in Arts & Sciences

AI/ML Intern Team Leader

Champaign, Illinois

January 2024 - December 2024

- Lead a team of 5 interns and collaborated closely with 3 managers in weekly meetings
- Built ML models with 100,000+ data points using Python and advanced techniques like XGBoost and BERT

### KM Internacional S.A.

Intern

Managua, Nicaragua

May 2023 - August 2023

- Oversaw comprehensive inventory management and streamlined administrative tasks for a small textile company
- Lead a team of three workers to achieve a 98% accuracy rate in record-keeping and a 15% reduction in operational costs

## PROJECTS/AWARDS

### Graduation Prediction Model

September 2024 - December 2024

- Increased AUC score from 0.65 to 0.82 by reducing noise in the dataset through feature engineering and outlier removal across 50,000+ student records
- Built a Streamlit interface with 12 key predictors for real-time predictions for university administrators

### 3rd Place, 2024 CIC Student Paper Challenge (Columbia University)

September 2024

- Analyzed the socioeconomic and mental health impacts of COVID-19 in Mexico, identifying that 42% of females and 40% of young adults faced job loss
- Proposed interventions to support vulnerable groups through geospatial and statistical analysis

### Paraphrase Detection

July 2024

- Achieved 90%+ accuracy in paraphrase detection using BERT, RoBERTa, and SBERT models on the MRPC dataset
- Developed a Gradio interface for real-time detection for easier evaluation of textual similarities

## SKILLS

Languages: English (Fluent), Spanish (Native), Korean (Native), French (Basic), Chinese (Basic)

Computer: Python (Advanced): Pandas, NumPy, Scikit-learn, TensorFlow, NLTK, Matplotlib, Seaborn, BERT, Random Forest, XGBoost, Gradio  
R (Intermediate): Tidyverse, ggplot2, dplyr, caret

Git (Intermediate), SAS (Intermediate), Tableau (Basic), SQL (Basic), Office (Advanced)

Certificates: Data Science DISCOVERY, Data Science Certificate, Coursera Business Analytics Certificate

## COMMUNITY SERVICE AND EXTRACURRICULAR ACTIVITIES

### Consultant in Illinois Business Consulting

September 2024 - December 2024

- Collaborating with interdisciplinary teams to deliver strategic solutions for real-world business challenges, including market analysis, operational optimization, and growth strategy, for clients ranging from Fortune 100 companies to nonprofits